

Homework - Catering Package Order Calculator Using Programmer Defined Functions

What to Submit?

You must submit the following **four files within a zip file** for this homework:

1. Completed Main Script

- Filename: HW_MAIN_PDF_Basu.m
- Complete the starter main script provided with the assignment

2. Create three programmer defined function Files

Academic Integrity

This homework must be completed **individually**. You may discuss general problem-solving strategies with others or seek help during student hours and from A², but MATLAB code must be your own work.

Problem Statement

A catering company offers the following packages (see table) for events. Each package includes a selection of food, serves a specified number of guests, and has a fixed price. The company needs a MATLAB application that helps a customer determine how many packages to order and the total cost of the order.

The completed application must calculate:

- The number of catering packages needed for an event
- The total cost after adding 10% tax
- The total number of guests the order can serve
- The number of unused servings

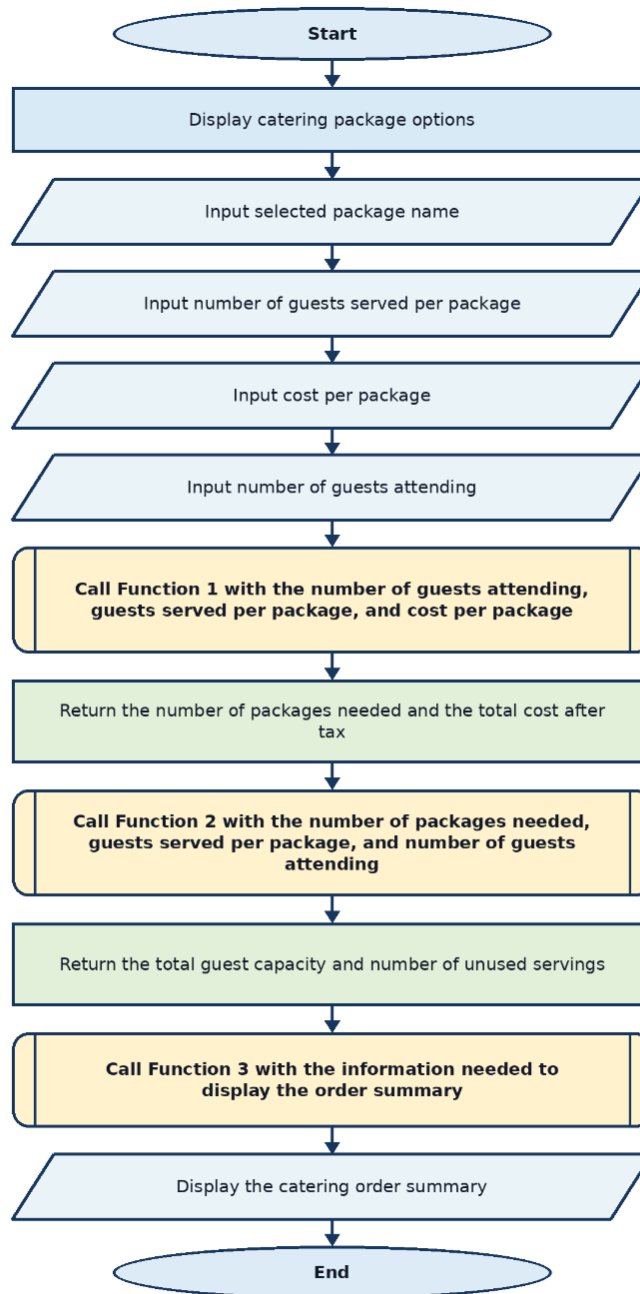
The application must be organized into *one main script and three reusable programmer-defined functions*. It should follow the below flowchart.

Catering Package Information:

Package	Guests Served	Price
Classic Lunch Package	10	\$150.00
Deluxe Dinner Package	15	\$300.00
Appetizer Package	20	\$180.00

Flowchart:

Catering Package Order Calculator



Main Script Requirements

Complete the provided HW_MAIN_PDF_Basu.m starter main script file. Do the following:

- Rename the main file using your last name.
- Update the provided header with your information.
- Keep the provided catering package information unchanged.
- Prompt the customer to enter:
 - The selected package name
 - The number of guests served by the selected package
 - The price of the selected package
 - The number of guests attending the event
- Call the first programmer-defined function to calculate the number of packages needed and the total cost, including tax.
- Call the second programmer-defined function to calculate the total guest capacity and the number of unused servings.
- Call the third programmer-defined function to display the final order summary (check the below sample run).
- Follow the sequence shown in the provided flowchart.
- Use variables in all calculations.
- Include meaningful comments and appropriate white space.

Programmer Defined Function Requirements

Create three programmer-defined functions. You must determine appropriate function names, input arguments, output arguments, and variable names based on the provided flowchart and the requirements below. Include comments in the function explaining the function's purpose.

Function	Information Received	Information Returned	Required Task
Function 1	Number of guests attending, number of guests served by one package, and cost of one package	Number of packages needed and total cost after tax	Use the ceil built-in function to determine the number of packages needed. Calculate the package subtotal and add 10% tax to determine the final cost.
Function 2	Number of packages needed, number of guests served by one package, and number of guests attending	Total guest capacity and number of unused servings	Calculate the number of guests that the purchased packages can serve and the number of unused servings.

Function	Information Received	Information Returned	Required Task
Function 3	All information needed to produce the final order summary	None	Use printf statements to display the selected package, number of guests, number of packages needed, guest capacity, unused servings, and total cost. This function should perform no calculations.

Sample Run:

Command Window

```
=====
          CATERING PACKAGE OPTIONS
=====
Package           Guests Served   Price
-----
Classic Lunch Package   10           $ 150.00
Deluxe Dinner Package   15           $ 300.00
Appetizer Package       20           $ 180.00
=====

Input the following information to order

-----
Enter the name of the catering package: Classic Lunch Package
Enter the number of guests served by that package: 10
Enter the cost of that package: $150
Enter the number of guests attending: 15

Catering Order Summary:
-----
Selected package: Classic Lunch Package
Number of guests: 15
Packages needed: 2
The order can serve: 20 guests
Unused servings: 5
Total order cost: $330.00
\\ |
```